Experiment No. 4

Open Loop Buck Type Switching Converter

Objectives:

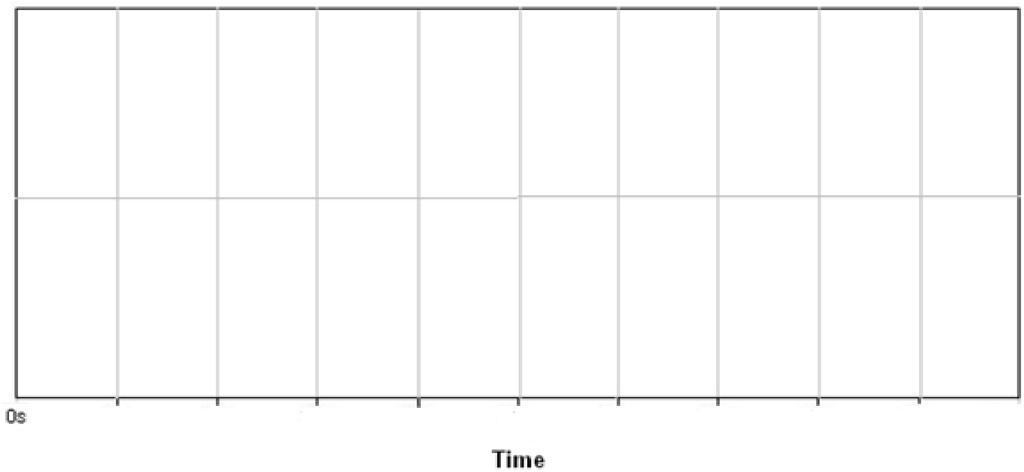
To become familiar with design of DC-DC converters and investigate the effect of inductor and capacitor on the performance of converter.

Procedure

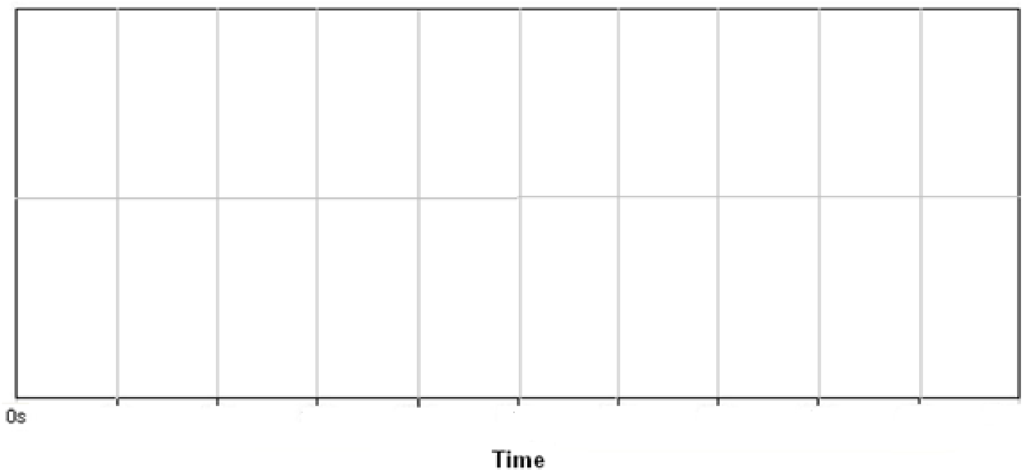
Design a 500W buck converter for an input voltage of 200VDC and output voltage of 120VDC. The switching frequency of your converter must be greater than 20kHz, whereas ripple current and voltage can be assumed to be 10% for the application. Simulate the designed buck converter in LTspice using a real switch and diode from well-known vendors of semiconductor devices. Make sure the switch you select is Silicon based MOSFET. You might need to design an ideal gate drive to make the transistor operate.

Section 1:

1. Plot inductor voltage and current for two switching cycles under steady-state conditions.



2. Plot switch and diode current for the same switching cycles used above.



Section 2: Determine the critical inductance for your converter and repeat the observations recorded in section 1.

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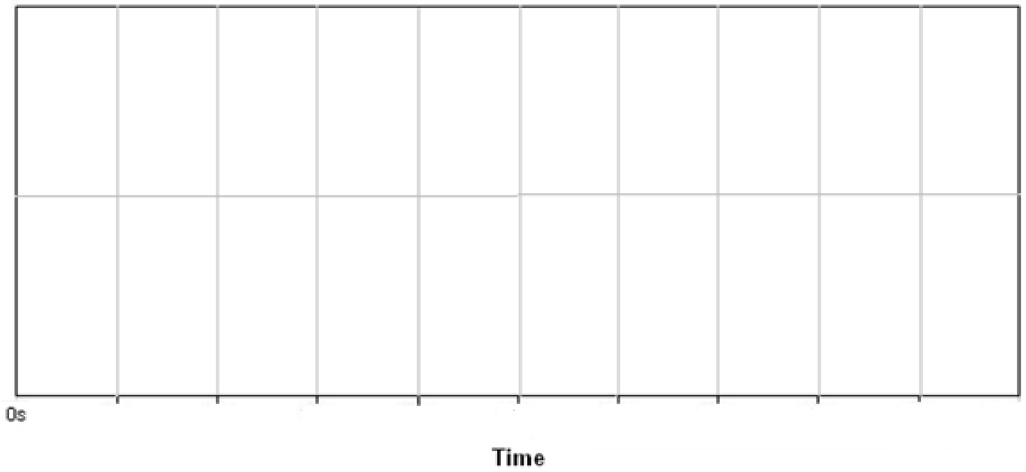

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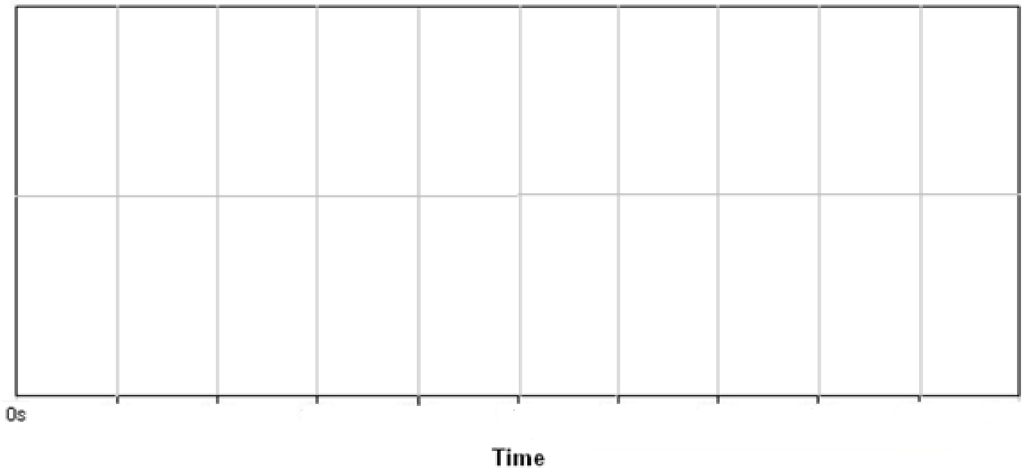

Section 3: Increase output load resistance in multiples of '2' and observe its impact on the above-mentioned quantities.

Section 4: Derive Fourier series expressions for input current for section 1 and 2 only. Match the theoretically calculated coefficients with your simulation-based results.